Project Management Plan

for

Project THEIA

Indoor Navigation Assistant for Blind and Visually Impaired Users

Version 1.2 (Final Phase I)

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Team EV — Cpt S 484, Fall 2026

October 18, 2026

Team repository: https://github.com/nemonto/cpts484-theia-ev

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Revision History

Name	Date	Reason for Changes	Version
Zichuo Wang	2026-09-17	Initial preliminary plan	1.0 draft 1
Zichuo Wang	2026-09-20	Week 4 schedule change	1.0 draft 2
Zichuo Wang	2026-09-21	Coordination update: co-authors, weekly reviews, W4 extended, no internal freeze	1.0 draft 3
Zichuo Wang	2026-10-17	Final Phase I version: D-3 owner and D-2 co-author reassigned, Peer Review moved to Oct 18, actual W5 schedule recorded	1.1
Zichuo Wang	2026-10-18	Justus Jones's late WRS §3 draft reviewed and merged into the WRS (v1.1); D-3 owner Justus Jones with co-author Zichuo Wang; D-5 co-author Young Kim	1.2

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1. Overview

This plan covers Phase I of the Cpt_S 484 team project for Project THEIA. It states what the team will produce, who owns each work product, how the team is organized and communicates, the risks it expects and how it will control them, and the schedule it will work to. This is the final Phase I version of the plan, updated to record what actually happened: roles in §2.3, risks in §3.3, and the schedule in §5.

Phase I key dates

Date	Milestone
Sun, Sep 20, 2026	Preliminary Plan Submission
Fri, Oct 16, 2026	Presentation video uploaded; all other work products complete
Sat, Oct 17, 2026	Final assembly of all work products
Sun, Oct 18, 2026	Phase I Final Submission and Peer Review form

1.1. Project Purpose, Objectives, and Success Criteria

THEIA is a smartphone application that helps blind and visually impaired people navigate indoors — from one location in a building to another location in the same building or a connected one, for example from a classroom to an office, a lab, or a restroom.

The purpose of this project, for CptS 484, is not to fully develop that application. It is an exercise in software requirements engineering. The team's objective for Phase I is to take the informal *THEIA Preliminary Definition* document, identify what is incomplete, inconsistent, ambiguous, redundant, or unsound in it, and produce a defensible set of requirements models and specifications with traceability from problems through to requirement specifications.

Objectives for Phase I

1. Identify and document issues in all three sections of the Preliminary Definition (domain and stakeholders, functional requirements, non-functional requirements), together with the design rationale for how the team resolves each one.
2. Produce a WRS document that traces Problems → Goals → Objectives → Requirement Specifications, for both functional and non-functional requirements.
3. Present four pairs of AS-IS/TO-BE scenarios (the template's Stevie pair plus three of the team's own) covering different aspects of the application, nominate a Top Priority TO-BE scenario, and analyze it with the Lesson 6 questions.
4. Produce a prototype mock-up and a preliminary user manual consistent with those TO-BE scenarios.
5. Document the team's use of generative AI and the team's own review of that output.

Success criteria

#	Criterion	How it is judged
SC-1	All six Phase I work products (D-1 to D-6) submitted by the deadline	Submitted on Canvas and present in the team repository
SC-2	Requirements are traceable end to end	Every Goal traces backward to a Problem and forward to an Objective; every Requirement Specification traces to an Objective

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#	Criterion	How it is judged
SC-3	Scenarios are genuinely distinct	The four AS-IS/TO-BE pairs (the template's Stevie pair plus three of the team's own) cover different aspects of the app, not variations of one
SC-4	Documents read as one coherent voice	Every member contributes to and reviews each work product at every stage; no section reads as an isolated hand-off
SC-5	No unreviewed AI output ships	Every AI-assisted passage is edited and verified by the owning member and checked by a second member

1.2. Project Deliverables

These are the required Phase I work products. No hypothetical industry artifacts (design models, test plans, deployment documents) are included, per the project specification.

ID	Deliverable	Weight	Delivery date	Method
D-0	Preliminary Project Plan + repository link	Not graded	Sun, Sep 20, 2026	Repo + Canvas
D-1	AS-IS/TO-BE scenarios presentation — slides + recorded 10–15 min video	20%	Sun, Oct 18, 2026	Slides in repo; YouTube link on Canvas
D-2	WRS document, Section 2 — Issues in the Preliminary Definition	25%	Sun, Oct 18, 2026	Repo + Canvas
D-3	WRS document, Section 3 — Improved Understanding (WRS model + traceability)	25%	Sun, Oct 18, 2026	Repo + Canvas
D-4	Prototype mock-up + preliminary User Manual	10%	Sun, Oct 18, 2026	Repo + Canvas
D-5	Project Plan, final Phase I version (this document)	10%	Sun, Oct 18, 2026	Repo + Canvas
D-6	AI Usage and Teamwork Report	10%	Sun, Oct 18, 2026	Repo + Canvas
D-7	Peer Review form — contribution evaluation of every team member, filled out and submitted individually by each member	Mandatory; grade held "incomplete" until submitted	Sun, Oct 18, 2026	Single file per member, uploaded to the instructor; file name includes the member's own name

1.3. Assumptions, Dependencies, and Constraints

ID	Statement
AS-1	The team scopes the indoor domain using the World Assumptions Masterlist v1.0 as the baseline, and records any assumption it adds or removes.
AS-2	No functioning software is required in Phase I; the prototype is a mock-up (sketched UI, concept drawing, or a non-functional GUI walkthrough).

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ID	Statement
AS-3	Where real blind or visually impaired users cannot be interviewed within the schedule, the team reasons from the position of a potential user and records that this is what it did.
AS-4	Every member owns or co-authors work products and reviews the others at every stage.
DE-1	D-3 depends on D-2: the Improved Understanding section is built from the issues and design rationale identified in the Issues section.
DE-2	D-1 and D-4 depend on the scenario set and requirements agreed in D-2/D-3; the user manual must instruct users through the TO-BE scenarios.
DE-3	D-6 depends on every member logging AI usage evidence continuously from the beginning.
DE-4	The team depends on course-provided templates.
CO-1	Course deadlines are fixed and cannot be renegotiated.
CO-2	The course is fully online; the team is distributed and works asynchronously. All five members carry a full course load, so effort is capped at roughly 4–6 hours per member per week.
CO-3	The presentation video must be 10–15 minutes and hosted as an unlisted YouTube video.
CO-4	Budget, procurement, and staffing cost — Not applicable to a course project.

1.4. References

1. Cpt S 484 Team Project Specification — Phase I, v2.0 (course materials).
2. Cpt S 484 Project THEIA — Preliminary Definition, v1.0 (course materials).
3. CPTS 484 FAQ (course materials).
4. World Assumptions Masterlist, v1.0 (course materials).
5. WRS-template.rtf and WRS-Sample-Incomplete.docx (course materials).
6. Theia-AS-IS_and_TO-BE_v2.0.pptx — presentation template (course materials).
7. Requirements Review Checklist.docx (course materials).

1.5. Definitions and Acronyms

Term	Definition
THEIA	The working name of the indoor navigation application (Theia, Greek goddess of vision).
WRS	World, Requirement, Specification — the requirements model relating the domain (World) to the system description (Requirement, Specification).
AS-IS scenario	A description of how a task is accomplished today, without the proposed system.
TO-BE scenario	A description of how the same task is accomplished with the proposed system.
Problem	A condition in the world today that the project intends to remove or reduce.
Goal	The state to be reached once a Problem is reasonably solved.
Objective	A functional or non-functional aim, derived from a Goal, that the app itself can address.
Requirement Specification (RS)	The lowest level of detail: a specific, verifiable feature of the app addressing an Objective.
Traceability	An explicit, recorded link between Problems, Goals, Objectives, and Requirement Specifications, in both directions.

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Term	Definition
Liaison	The single team member responsible for communication with the instructor.
Gen AI	Generative AI tools used in preparing deliverables, whose usage must be disclosed.

2. Project Organization

2.1. Process Model

The team follows the spiral requirements engineering process (Kotonya & Sommerville), which matches the structure of Phase I: the team starts from an informal statement of requirements and cycles through elicitation, analysis and negotiation, documentation, and validation, producing a progressively better requirements document on each pass.

Cycle	Activity	Produces
1	**Elicitation** — read the Preliminary Definition and the World Assumptions Masterlist; each member logs candidate issues and missing requirements	Raw issues list
1	**Analysis and negotiation** — the team consolidates duplicate issues, resolves disagreements, and records design rationale for each resolution	WRS §2 Issues (D-2)
2	**Documentation** — Problems and Goals are derived from the resolved issues, decomposed into functional and non-functional Objectives, then into Requirement Specifications, with traceability tables	WRS §3 Improved Understanding (D-3)
2	**Documentation — the agreed requirements are expressed as four AS-IS/TO-BE scenario pairs, a prototype mock-up, and a user manual**	D-1, D-4
3	**Validation** — at the end of every week, the team reviews that week's output against the *Requirements Review Checklist* before the next cycle builds on it; the last pass checks the assembled documents for consistency across work products	Review notes; revised documents
3	**Decision point** — accept the document set, or re-enter the spiral on the sections that failed review	Phase I submission

2.2. Organizational Structure

The team is five members. Each work product has an owner and a co-author, and every member is a reviewer of every work product at each stage.

- **Team Liaison:** Zichuo Wang. Posts all instructor communication to the team within 24 hours.
- **Repository maintainer:** Zichuo Wang. Owns the repo structure and merges; any member commits to their own work-product folder directly.
- **All members:** requirements analysts. All five participate in identifying issues and in reviewing every work product.

Communication and collaboration

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Channel	Use
Discord	Primary channel. Async text by default; per-work-product channels.
GitHub	Version control and single source of truth for every document.
Canvas / email	Instructor communication.

2.3. Roles and Responsibilities

Task distribution follows a **one owner, one co-author, four reviewers** model: each work product has one accountable owner and a co-author who writes it with the owner, and the other four members (including the co-author) review it every week, not only before submission.

Work product	Owner (point person)	Co-author	Reviewers
D-1 AS-IS/TO-BE presentation + video	James Zuniga	Austin Aragon	All
D-2 WRS §2 — Issues	Austin Aragon	James Zuniga	All
D-3 WRS §3 — Improved Understanding + traceability	Justus Jones	Zichuo Wang	All
D-4 Prototype mock-up + User Manual	Young Kim	Zichuo Wang	All
D-5 Project Plan (this document)	Zichuo Wang	Young Kim	All
D-6 AI Usage and Teamwork Report	Zichuo Wang	Young Kim	All

Role changes in W5. Justus Jones, the D-3 owner, did not reply to the team during the final week, so Zichuo Wang completed WRS §3 for the assembly on Sat, Oct 17. On Sun, Oct 18, Justus submitted an independent §3 draft and his AI usage log (PR #1). The team checked the draft against the slides, the prototype, and the manual, and merged its additions into WRS v1.1: priorities for every requirement, acceptance criteria for the non-functional requirements, error handling, and the Preliminary Definition coverage table. Parts that conflicted with the recorded scenarios or the prototype were not adopted.

Owner responsibilities. Lead the work product and pull the team's input into it, keep the draft in the repository, raise blockers in Discord early, incorporate review comments, and bring that week's progress to each weekly review.

Co-author responsibilities. Write an agreed part of the work product with the owner. Because the co-author knows the work product from the inside, they can take it over if the owner falls behind.

Reviewer responsibilities. At the end of every week, review what each work product produced that week against the Requirements Review Checklist and leave comments in the repository or the relevant Discord channel. Comments are resolved before the next week's work builds on them. Reviewing is scheduled work, not goodwill — it appears in every week of Section 5, so no work product is reviewed only once at the end.

Shared responsibility — the issues list. Every member contributes at least three candidate issues from the Preliminary Definition during the elicitation week (§5, W2). This is what keeps D-2 from being one person's opinion.

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3. Managerial Process Plans

3.1. Management Objectives and Priorities

The priority order for Phase I is completeness of the required work products first, depth second. A submitted set of six coherent work products with traceability outscores three excellent documents and three missing ones.

Progress is managed by three mechanisms:

1. **Owner accountability.** Every work product has one owner and one co-author, and the whole team reviews it. "The team" is never the owner of anything, but the whole team is involved in everything.
2. **Draft-early visibility.** Work is committed to the repository in progress, so slippage is visible in the commit history rather than discovered at the deadline.
3. **Review at every stage.** Each week ends with a team review of that week's output at the Sunday checkpoint. Problems are caught while the work is in progress, not in a final review window.

3.2. Assumptions, Dependencies, and Constraints

See Section 1.3. The dependency with the most schedule risk is DE-1 (D-3 cannot be finished before D-2 stabilizes). This is managed by fixing an internal deadline for D-2 of Sun, Oct 4 — two weeks ahead of the final submission — and by having the D-3 owner and co-author start on Problems and Goals in that same week, working from the draft issues list rather than waiting for D-2 to be final.

3.3. Risk Management

#	Risk	Type	Likelihood / Impact	Description
R-1	Uneven member participation	Managerial	Unlikely / High	One or more members do not respond or do not deliver, leaving a work product unowned close to the deadline.
R-2	Dependency stall on the WRS	Technical	Likely / Medium	D-3 stalls because the issues in D-2 are not settled, compressing traceability work into the final week.
R-3	Scope creep in requirements	Technical	Likely / Medium	The team keeps adding requirements to an open-ended problem and never converges. The FAQ warns against an "abundant laundry list."
R-4	Inconsistent voice across work products	Managerial	Likely / Medium	Documents read as five separate submissions; the specification explicitly warns against isolated work.

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#	Risk	Type	Likelihood / Impact	Description
R-5	Unreviewed or hallucinated AI output	Technical	Unlikely / High	AI-generated content is submitted without verification; the specification states grades are negatively impacted by errors and hallucinations.
R-6	Missing AI usage evidence	Managerial	Likely / Medium	Members do not keep chat logs or screenshots, and D-6 cannot be evidenced at the end of the phase.
R-7	Presentation recording runs late	Managerial	Unlikely / High	Slides finish on time but the 10–15 minute recording and upload slip past the deadline.
R-8	Loss of work	Technical	Unlikely / High	Work kept only on a local machine is lost to a device failure.
R-9	Peer Review form not submitted	Managerial	Unlikely / High	Peer grading is mandatory and the course grade is held "incomplete" until the form is received.
R-10	Contribution allocation disputed	Managerial	Unlikely / Medium	Members disagree on the D-7 percentages because there is no shared record of who produced what.

Monitoring and control

#	Monitoring and controlling mechanism
R-1	Weekly checkpoint in Discord; every work product has a co-author who already works on it and can take it over. If an owner misses two consecutive checkpoints, the co-author takes over. Outcome: R-1 occurred in W5. The D-3 owner did not reply during the final week, so the D-3 co-author completed WRS §3 for the assembly, and James Zuniga took the D-2 co-author role. The owner's draft arrived after the assembly (Oct 18); after review, its additions were merged into WRS v1.1 (see §2.3).
R-2	Internal deadline for D-2 set two weeks before the phase deadline; D-3 starts from the draft issues list. The D-2 and D-3 owners and co-authors share the #wrs channel and post progress through weeks 3–5.
R-3	Scope is bounded by the World Assumptions Masterlist baseline recorded under AS-1. The team targets a small set of crucial requirements covering the three angles the FAQ names — misleading, missing, and implied — rather than exhaustive coverage.
R-4	Weekly team review of every work product (§5) using the *Requirements Review Checklist*; the Liaison does a single-voice editing pass across all documents in W5 before submission.
R-5	Every AI-assisted passage is verified by its owner against the source documents and checked by one reviewer; the verification method is written up in D-6 as the specification requires.
R-6	Evidence is committed to 05-ai-usage-teamwork/ continuously.
R-7	Recording is scheduled as its own dated task (video uploaded by Fri, Oct 16), not as the tail of slide preparation.
R-8	All work lives in the GitHub repository. Nothing is submitted that exists only on one machine.

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#	Monitoring and controlling mechanism
R-9	D-7 is listed in §1.2 and in the schedule from the start of the phase, pinned in the Discord reference message, and raised at the weekly checkpoints. Its due date moved from Mon, Oct 12 to Sun, Oct 18; the Liaison reminds the team once before Oct 18, and each member remains responsible for their own submission.
R-10	The repository commit history, the per-work-product ownership in §2.3, and the weekly written checkpoints form a shared, factual record of contribution that any member can consult before filling out D-7. The record is built continuously rather than reconstructed, which is what makes it credible.

4. Technical Process Plans

4.1. Methods, Tools, and Techniques

Purpose	Tool
Requirements method	WRS model; AS-IS/TO-BE scenario analysis; question-based analysis of the Top Priority TO-BE scenario
Document authoring	Course-provided templates (WRS-template.rtf, ProjectPlanTemplate_updated.docx, Theia-AS-IS_and_TO-BE_v2.0.pptx); Google Drive for concurrent drafting, exported to the repository
Version control	GitHub — single source of truth for all work products
Communication	Discord; Canvas and email for instructor contact
Prototype	Sketched or wireframed UI mock-up (e.g. Figma, or hand-drawn concept drawings), with explicit attention to what a GUI means for a user who cannot see the screen — voice interaction, audio feedback, haptics
Presentation	Slides from the provided template; screen-recorded narration; unlisted YouTube upload
Gen AI	ChatGPT, Gemini / NotebookLM, and Claude / Claude Code. Used for drafting, restructuring, critique, image mock-ups, and final assembly, with usage logged; all output verified by the owning member and one reviewer before it enters a deliverable

No implementation technology is selected in Phase I. The specification states no coding is required, and choosing a stack now would be a requirements decision made for the wrong reason.

4.2. Software Documentation

Phase I produces requirements documentation only. The document set is: the WRS document (issues, improved understanding, traceability tables), the AS-IS/TO-BE presentation, the prototype and preliminary user manual, this project plan, and the AI usage and teamwork report.

Every document is versioned in the repository. This plan is maintained and updated throughout Phase I; a final version is submitted with the Phase I deliverables, with changes recorded in the Revision History.

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5. Work Elements and Schedule

Week	Dates	Work elements	Owner(s)	Milestone
W1	Sep 14 – Sep 20	Team formation; Discord and GitHub set up; preliminary plan drafted and reviewed; roles assigned	Zichuo Wang (all review)	M1: Preliminary Plan + repo link submitted — Sun, Sep 20
W2	Sep 21 – Sep 27	All members read the Preliminary Definition and the World Assumptions Masterlist; each logs ≥3 candidate issues; team consolidates the issues list and records scope assumptions (AS-1). In parallel, candidate AS-IS/TO-BE scenarios are proposed and the team's three pairs are selected to go with the template's Stevie pair — pulled forward into W2 because D-1 and D-4 both depend on them. Team review of the issues list and scenario choice at the Sep 27 checkpoint	All; consolidated by Austin Aragon + James Zuniga; scenarios by James Zuniga + Austin Aragon	M2: Consolidated issues list, scope assumptions, and scenario pairs chosen — Sun, Sep 27
W3	Sep 28 – Oct 4	WRS §2 complete: issues in domain/stakeholders/objectives (2.1), functional requirements (2.2), non-functional requirements (2.3), each with design rationale. WRS §3 started: Problems (3.1.1) and Goals (3.1.2) derived from the resolved issues. Slide skeleton built from the provided template; prototype mock-up sketched; Top Priority TO-BE scenario selected. Team review of WRS §2 and the Problems/Goals draft at the Oct 4 checkpoint, before §3 builds on them	Austin Aragon + James Zuniga (D-2); Justus Jones + Zichuo Wang (D-3); James Zuniga + Austin Aragon (D-1); Young Kim + Zichuo Wang (D-4); all review	M3: WRS §2 complete; Problems and Goals drafted — Sun, Oct 4

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Week	Dates	Work elements	Owner(s)	Milestone
W4–W5	Oct 5 – Oct 18 (midterm week, extended through W5)	WRS §3 completed: improved understanding (3.1.3), functional and non-functional requirement specifications (3.2.1, 3.2.2), and traceability tables in the WRS-Sample format. Slides finished; question-based analysis of the Top Priority scenario; comparison with existing aids (guide dog, cane). User manual drafted against the TO-BE scenarios. Midterm week: this work runs through W5 instead of ending Oct 11. Each work product is reviewed by the team as its sections are drafted — at the Oct 11 checkpoint and as sections finish in W5	Justus Jones + Zichuo Wang (D-3); James Zuniga + Austin Aragon (D-1); Young Kim + Zichuo Wang (D-4); all review	M4: Midterm-week checkpoint — Sun, Oct 11 (progress reviewed by the team)
W5	Oct 13 – Oct 18	Remaining W4 work finished and reviewed. Presentation recorded (10–15 min) and uploaded to YouTube as unlisted. All other work products (slides, WRS §2, prototype mock-up, user manual, AI usage evidence) completed by Fri, Oct 16. Sat, Oct 17: Zichuo Wang assembles all work products into their final form, completes WRS §3 as the D-3 co-author, runs the single-voice editing pass and the cross-product consistency check, finalizes this plan, and builds the WRS document and the AI Usage and Teamwork Report. Sun, Oct 18: Justus Jones's §3 draft (PR #1) reviewed and its additions merged into WRS v1.1; submission	All; Zichuo Wang assembles and runs the editing pass (Sat, Oct 17); Zichuo Wang (D-3 co-author); Justus Jones (D-3 draft, Oct 18); Zichuo Wang + Young Kim (D-5 final version, D-6)	M5: Video uploaded and all other work complete — Fri, Oct 16 · Assembly — Sat, Oct 17 · Phase I submission — Sun, Oct 18
—	Sun, Oct 18	Peer Review form submitted — each member completes one form per team member including themselves, allocates contribution percentages summing to 100%, and uploads a single file named with their own name	All, individually	M6: Peer Review submitted — Sun, Oct 18

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